

Haodong Yan

☎ (+86) 15829011780 | @ hyan758@connect.hkust-gz.edu.cn

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Education

The Hong Kong University of Science and Technology (Guangzhou)

Ph.D. in Intelligent Transportation

Guangzhou, China

Sep. 2024 – Present

Xi'an Jiaotong University

Master of Mechanical Engineering

Rank: 1/281

GPA: 3.73/4.0, Average Score: 89.68

Xi'an, China

Sep. 2021 – Jul. 2024

Xi'an Jiaotong University

Bachelor of Mechanical Engineering

Xi'an, China

Sep. 2017 – Jan. 2021

GPA: 3.75/4.3, Average Score: 88.75

Xi'an Jiaotong University

Second Bachelor of Computer Science and Technology

Xi'an, China

Jul. 2019 – Jul. 2021

Selected Publications

- [1] **Haodong Yan**, Zhide Zhong, Jiaguan Zhu, Junjie He, Weilin Yuan, Wenxuan Song, Xin Gong, Yingjie Cai, Guanyi Zhao, Xu Yan, Bingbing Liu, Ying-Cong Chen, Haoang Li, “S-VAM: Shortcut Video-Action Model by Self-Distilling Geometric and Semantic Foresight,” *ECCV*, 2026.
- [2] **Haodong Yan***, Hang Yu*, Zhide Zhong*, Weilin Yuan, Xin Gong, Zehang Luo, Chengxi Heyu, Junfeng Li, Wenxuan Song, Shunbo Zhou, Haoang Li, “Open-world Hand-Object Interaction Video Generation Based on Structure and Contact-aware Representation,” *CVPR*, 2026.
- [3] Zhide Zhong*, **Haodong Yan***, Junfeng Li, Xiangchen Liu, Xin Gong, Tianran Zhang, Wenxuan Song, Jiayi Chen, Xinhua Zheng, Hesheng Wang, Haoang Li, “FlowVLA: Visual Chain of Thought-based Motion Reasoning for Vision-Language-Action Models,” Preprint, under review at *IEEE Transactions on Robotics*.
- [4] **Haodong Yan**, Pengxu Hou, Zhide Zhong, Xinhua Zheng, Zhe Liu, Hesheng Wang, Haoang Li, “CARE: Contextually-Aligned and Realistic 4D Scene Generation from a Single Image and Text,” Preprint, under review at *IEEE Transactions on Image Processing*.
- [5] **Haodong Yan**, Jiaguan Zhu, Mingyuan Jia, Ruiqing Yin, Junjie He, Zhide Zhong, Junfeng Li, Jinxuan Lu, Hengtao Li, Tianran Zhang, Jiayi Chen, Wenxuan Song, Wen Chen, Yuxiang Gao, Haoang Li, “PSG-JEPA: Is Forward Prediction Enough? Physical State Grounding for JEPA World Models,” *arXiv*, 2026.
- [6] **Haodong Yan**, Junfeng Li, Junjie He, Zhide Zhong, MingMing Yu, Wenxuan Song, Jiaguan Zhu, Yangyang Zheng, Yuqiao Du, Jiadi You, Yingjie Cai, Xu Yan, Guanyi Zhao, Bingbing Liu, Haoang Li, “Robust-WAM: Bridging Generative Pretraining and Semantic Foresight in World-Action Models,” *arXiv*, 2026.
- [7] Zhide Zhong*, **Haodong Yan***, Junfeng Li, Junjie He, Tianran Zhang, Haoang Li, “VLA-OPD: Bridging Offline SFT and Online RL for Vision-Language-Action Models via On-Policy Distillation,” *arXiv*, 2026.
- [8] Zhide Zhong, Junfeng Li, Junjie He, **Haodong Yan**, Xin Gong, Guanyi Zhao, Yingjie Cai, Jiantao Gao, Xu Yan, Bingbing Liu, Yingcong Chen, Liuqing Yang, Haoang Li, “DualCoT-VLA: Visual-Linguistic Chain of Thought via Parallel Reasoning for Vision-Language-Action Models,” *arXiv*, 2026.
- [9] Huashuo Lei, Wenxuan Song, Huarui Zhang, Jieyuan Pei, Jiayi Chen, **Haodong Yan**, Han Zhao, Pengxiang Ding, Zhipeng Zhang, Lida Huang, Donglin Wang, Yan Wang, Haoang Li, “RoboMemArena: A Comprehensive and Challenging Robotic Memory Benchmark,” *arXiv*, 2026.

- [10] Junfeng Li, Junjie He, Zhide Zhong, Yangyang Zheng, Pingyue Sheng, Jiayu Dong, Ruixin Li, **Haodong Yan**, Jiaguan Zhu, Tianran Zhang, Runze Yu, Wen Chen, Liuqing Yang, Yuxiang Gao, Haoang Li, “DyPES-VLA: Learning Shared Dynamics Priors and Embodiment-Specific Control for Cross-Embodiment Manipulation,” *arXiv*, 2026.
- [11] Junjie He, Junfeng Li, Zhide Zhong, **Haodong Yan**, Ruixin Li, Yangyang Zheng, Jiaguan Zhu, Tianran Zhang, Yuqiao Du, Wen Chen, Shunbo Zhou, Haoang Li, “SG-WAM: Text-Grounded and Spatial-aware Semantic Guidance for World-Action Models,” *arXiv*, 2026.
- [12] GigaWorld Team (incl. **Haodong Yan**), “GigaWorld-Policy-0.5: A Faster and Stronger WAM Empowered by AutoResearch,” *arXiv*, 2026.
- [13] Wenxuan Song, Jiayi Chen, Xiaoquan Sun, Huashuo Lei, Yikai Qin, Wei Zhao, Pengxiang Ding, Han Zhao, Tongxin Wang, Pengxu Hou, Zhide Zhong, **Haodong Yan**, Donglin Wang, Jun Ma, Haoang Li, “Rethinking the Practicality of Vision-language-action Model: A Comprehensive Benchmark and An Improved Baseline,” *ICRA*, 2026.
- [14] Wenxuan Song, Ziyang Zhou, Han Zhao, Jiayi Chen, Pengxiang Ding, **Haodong Yan**, Yuxin Huang, Feilong Tang, Donglin Wang, Haoang Li, “ReconVLA: Reconstructive Vision-Language-Action Model as Effective Robot Perceiver,” *AAAI*, 2026 (Outstanding Paper Award).
- [15] **Haodong Yan***, Zhiming Hu*, Syn Schmitt, Andreas Bulling, “GazeMoDiff: Gaze-guided Diffusion Model for Stochastic Human Motion Prediction,” *Pacific Graphics*, 2024.
- [16] Cheng Lei, Junpeng Hu, **Haodong Yan**, Mariia Gladkova, Tianyu Huang, Yun-Hui Liu, Daniel Cremers, Haoang Li, “Physically-Based Photometric Bundle Adjustment in Non-Lambertian Environments,” *IROS*, 2024.
- [17] **Haodong Yan**, Fudong Li, Jinglong Chen, Zijun Liu, Jun Wang, Yong Feng, Xinwei Zhang, “A Graph Embedded in Graph Framework with Dual-sequence Input for Efficient Anomaly Detection of Complex Equipment under Insufficient Samples,” *Reliability Engineering & System Safety*, 2023.
- [18] **Haodong Yan**, Zijun Liu, Jinglong Chen, Yong Feng, Jun Wang, “Memory-augmented Skip-connected Autoencoder for Unsupervised Anomaly Detection of Rocket Engines with Multi-source Fusion,” *ISA Transactions*, 2023.
- [19] **Haodong Yan**, Jun Wang, Jinglong Chen, Zijun Liu, Yong Feng, “Virtual Sensor-based Imputed Graph Attention Network for Anomaly Detection of Equipment with Incomplete Data,” *Journal of Manufacturing Systems*, 2022.

* equal contribution.

Work Experience

RobotEra

Internship

Beijing, China

Jul. 2026 – Present

- Vision-Language-Action and World Models for robot manipulation

GigaAI

Internship

Beijing, China

May 2026 – Jul. 2026

- World-Action Model (WAM) foundation model training (GigaWorldPolicy)

Institute for Visualisation and Interactive Systems, University of Stuttgart

Internship

Stuttgart, Germany

Jul. 2023 – Oct. 2023

- Code implementation
- Paper writing

Computer Vision Group, Technical University of Munich

Remote Internship

Remote

Apr. 2023 – Dec. 2023

- Code implementation
- Paper writing
- Data processing

Honours and Awards

National Scholarship – Ministry of Education, China	2023
National Scholarship – Ministry of Education, China	2022
China Optics Valley Scholarship – Wuhan High-tech Zone Government	2022
Outstanding Graduates – Xi'an Jiaotong University	2021
HIWIN Elite Scholarship – HIWIN Technologies Corp.	2021
National First Prize – Chinese University Students Mechanical Innovation Competition	2020
National First Prize – Chinese Intelligent Robot Fighting Competition	2019

Skills

Programming: Python (PyTorch, OpenCV, Transformers), C++, L^AT_EX, Git.

Research Interests: Robot Learning (Video-Action Models, World Models), Video Generation Models.

Languages

English: IELTS Overall 7.0 (Listening 7.0, Reading 8.5, Writing 6.5, Speaking 6.0).

Mandarin Chinese: Native.